

1. (10 points) Model the problem as a CSP, that is define variables, their domains and constraints between them. There are different ways of modelling this. Typically you want to have fewer variables and smaller domains (to reduce the size of the state space) and fewer or simpler constraints (to speed up constraint propagation). What are your variables and their domains?

Our variables are all the objects in the problem and the domain for each is the house numbers.

2. (10 points) How big is the state space? Shortly, explain your answer.
Our state space is 5^{25} because that is the number of states possible without constraints. There are 5 possible houses each variable can be in and 25 total variables in all.
3. (60 points) Download the material for the lab and implement your model (implement all things marked with TODO).
4. (20 points) Run the program and compare the results. How do you explain these different results? What does that tell you about the usefulness of different propagation algorithms and heuristics for this problem?

Our results are like the ones found on the Wikipedia page so there are no differences that we can see since everything is where it is supposed to be. We started off with a very different result but after tweaking the constraints a bit we managed to get the right results showing us the importance of understanding your heuristics and propagation algorithms well to utilize them correctly and to their full potential.